

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

เลขที่นั่งสอบ

Midterm Examination of Semester 2

Year: 2012

Subject: 392152 Chemistry 2

Section: 15-16

Date: 5 January 2013

Time: 13.00-15.00

Name: _____ ID: _____ Class: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. Only calculators are allowed in the examination.
4. Dictionaries are NOT allowed.
5. This exam is a closed book examination.
6. Electronic communication devices are NOT allowed in the examination room.
7. The examination has 6 pages (including this page), 2 sections and a total score of 60 points.
8. Write all your answers on the answer sheet.

WebElements: the periodic table on the world-wide web

<http://www.webelements.com/>

[illegible]

***lanthanoids**

**actinoids

lanthanum	57	cerium	58	praseodymium	59	neodymium	60	promethium	61	samarium	62	europlum	63	gadolinium	64	terbium	65	dysprosium	66	holmium	67	erbium	68	thulium	69	ytterbium	70
La	138.91	Ce	140.12	Pr	140.91	Nd	144.24	Pm	[145]	Sm	150.36	Eu	151.96	Gd	157.25	Tb	158.93	Dy	182.50	Ho	164.93	Er	167.26	Tm	168.93	Yb	173.04
actinium	89	thorium	90	protactinium	91	uranium	92	neptunium	93	plutonium	94	americium	95	curium	96	berkelium	97	californium	98	eksteinium	99	fermium	100	mendelevium	101	nobelium	102
Ac		Th	232.04	Pa	231.04	U	238.03	Np	[237]	Pu	[244]	Am	[243]	Cm	[247]	Bk	[247]	Cf	[251]	Es	[252]	Fm	[257]	Md	[258]	No	[259]

Symbols and names: the symbols and names of the elements, and their spellings are those recommended by the International Union of Pure and Applied Chemistry (IUPAC - <http://www.iupac.org/>). Names have yet to be proposed for the most recently discovered elements 110–112 and 114 so those used here are IUPAC's temporary systematic names. In the USA and some other countries, the spellings aluminium and caesium are normal while in the UK and elsewhere the common spelling is sulphur.

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Element labels: the numeric system (1–18) used here is the current IUPAC convention.

Atomic weights (mean relative masses): Apart from the heaviest elements, these are the IUPAC 2001 values and given to 5 significant figures. Elements for which the atomic weight is given within square brackets have no stable nuclides and are represented by the element's longest lived isotope.

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Part 1. Multiple Choice (22 points)

Identify the letter of the choice that best completes the statement or answers the question.

- ___ 1. Which of the following is not a common state of matter under ambient conditions?
a. solid b. liquid c. plasma d. gas
- ___ 2. A change in solid direct to gas is called _____.
a. sublimation b. melting c. evaporation d. condensation
- ___ 3. In which physical change is an exothermic process?
a. melting b. freezing c. evaporation d. sublimation
- ___ 4. Which of the following is true of the Celsius scale?
a. 373 degree is the boiling point of water.
b. 0 degree is the absolute zero.
c. 32 degree is the freezing point of water.
d. 0 degree is the melting point of water.
- ___ 5. The melting point of the metal iron in degrees Celsius is approximately _____.
a. -42 b. 0 c. 100 d. 1535
- ___ 6. What law may be stated qualitatively as "When you squeeze a gas, it gets smaller"?
a. Boyle's law b. Charles' law c. Ideal gas law d. Graham's Law
- ___ 7. The amount of a solute dissolved in a given amount of solvent is represented by the _____.
a. mass of the solution
b. volume of the solution
c. concentration of the solution
d. temperature of the solution
- ___ 8. The units of _____ are moles of solute per kg of solvent.
a. molality b. molarity c. mole fraction d. percentage by mass
- ___ 9. Tyndall effect is occurred only in _____.
a. solution b. suspension c. colloid d. compound
- ___ 10. A fluid's resistance to flow is its _____.
a. density b. viscosity c. pressure d. vaporization

11. Above sea level, water boils at a _____ temperature due to the _____ atmospheric pressure.
- a. higher, increase

b. lower, increase

c. higher, decrease

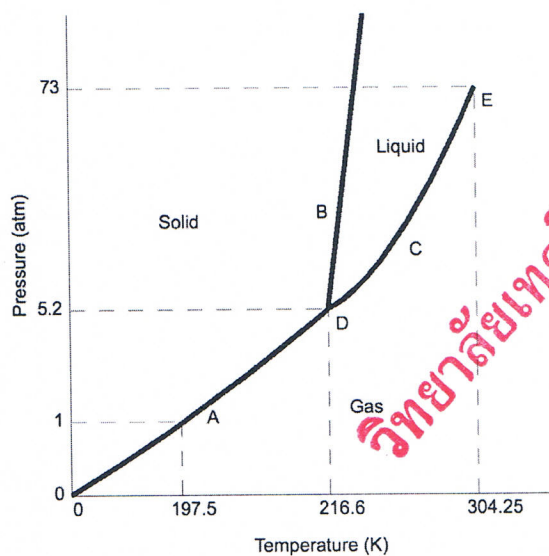
d. lower, decrease

Part 2 Answer (38 points)

1. Fill out the blanks. (6 points)

Changing	Changing of phase	Process	Endo/Exothermic
ice→water	solid→liquid	melting	endothermic
water→ice	_____	_____	_____
water→vapour	_____	_____	_____

2.

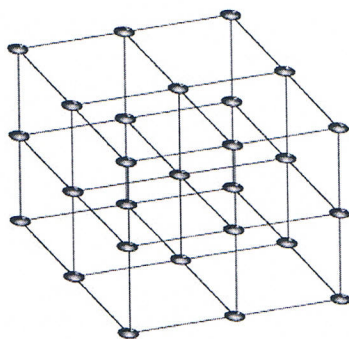


According to the phase diagram on the left:

(3 points)

- The critical point is _____.
- The tripple point is _____.
- The phase of this substance at the normal condition is _____.

3.



Draw the unit cell of the crystal on the left:

(3 points)

4. A 2.00-L sample of gas at 675 torr pressure is changed at constant temperature until its pressure is 925 torr. What is its new volume? (3 points)

5. A 22.5-mL sample of gas is warmed at constant pressure from 18°C to 36°C . What is its final volume? (3 points)

6. How many moles of SO_2 are present in a 765-mL sample at 37°C and 775 torr? (5 points)

7. Oxygen gas and sulfur dioxide (SO_2) gas are at the same temperature. What is the ratio of the effusion rates of their molecules? (4 points)

8. A solution is prepared by dissolving 36.5 g of gaseous HCl in 210 g of water to make 250 mL of solution. Calculate (6 points)
- the molality of this solution
 - the mole fraction of HCl and water in this solution

9. The concentration of a KBr solution is 1.5 M. What volume of the solution contains 30 g of KBr? (5 points)

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